

Tracing the Evolution of Oxygen Use on the Tree of Life

BIOLM0047

Genome Biology and Genomics analysis

MSc Bioinformatics

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1 Evaluate the quality of the raw sequencing data using FastQC.

As part of a project examining the evolution of oxygen use in metabolism across the Tree of Life, DNA from a novel aerobic microorganism was sequenced and sent for analysis. The quality of the data (GBG_2025_R1.fastq and GBG_2025_R2.fastq) was first assessed using FastQC v0.12.1 (Andrews, 2010). The FastQC report indicated that there was a clear drop in per base quality towards the end of the reads as well as discrepancies between theoretical and actual GC counts per read (Figure 1).

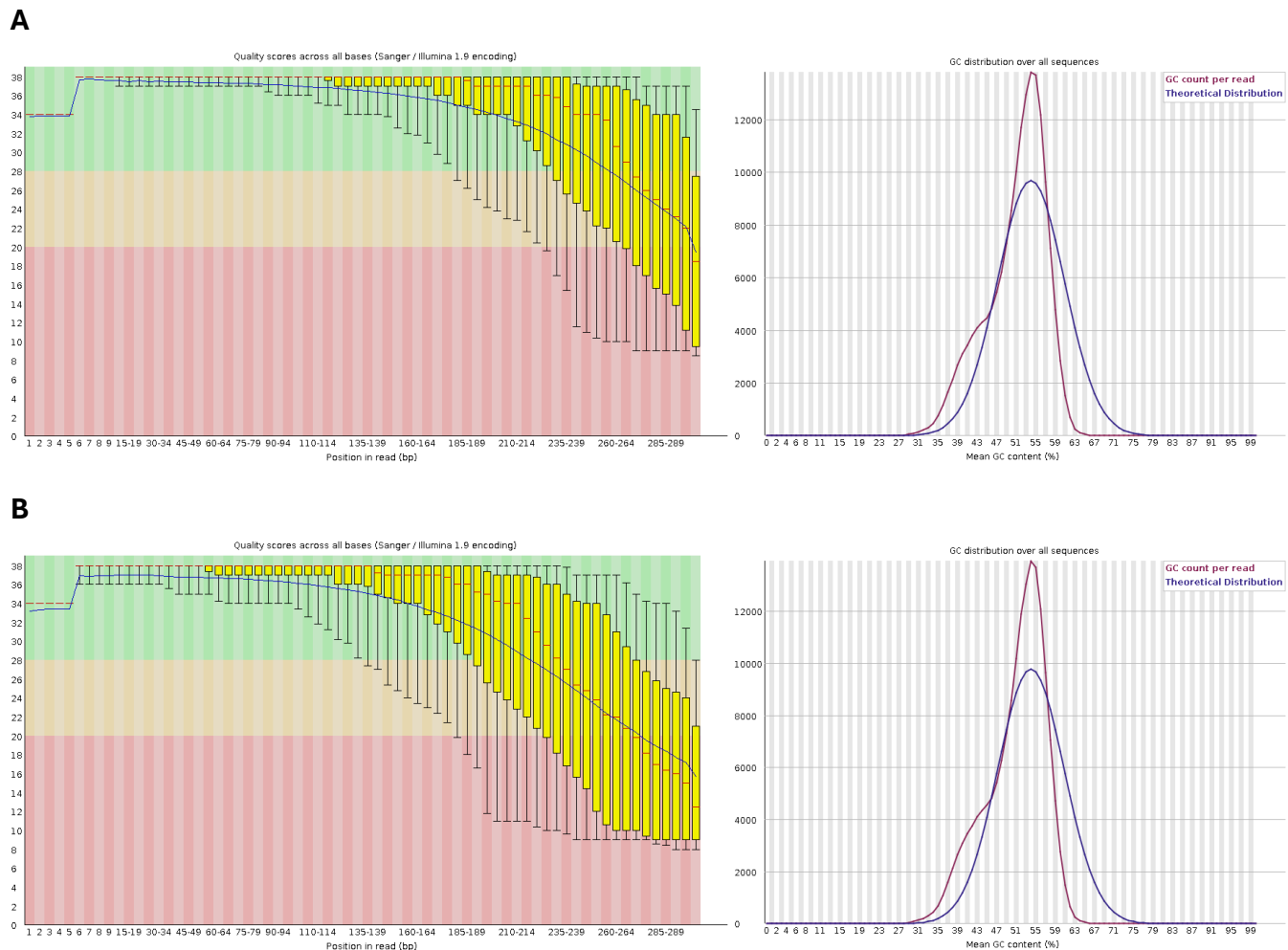


Figure 1. Pre-trimming FastQC reports. Per base sequence quality (left) and per sequence GC content (right).
A) GBG_2025_R1.fastq **B)** GBG_2025_R2.fastq

2 Trim the reads to remove adapter sequences and low-quality bases. Assemble the reads and indicate the total length of the assembled contigs and the N50.

To improve the quality of the data, sequences were trimmed using Trimmomatic v0.39 and fastp v0.23.4 and results compared with FastQC, revealing Trimmomatic to have performed slightly better with per base sequence quality (Figure 2) (Bolger *et al.*, 2014; Chen, 2023). Continuing with the Trimmomatic sequences, these were assembled using MEGAHIT v1.2.9, and statistics obtained with QUAST v5.2.0 (Gurevich *et al.*, 2013; Li *et al.*, 2016, 2015). Total length and N50 for the assembled contigs were 2,394,033 bp and 145,186 bp, respectively.

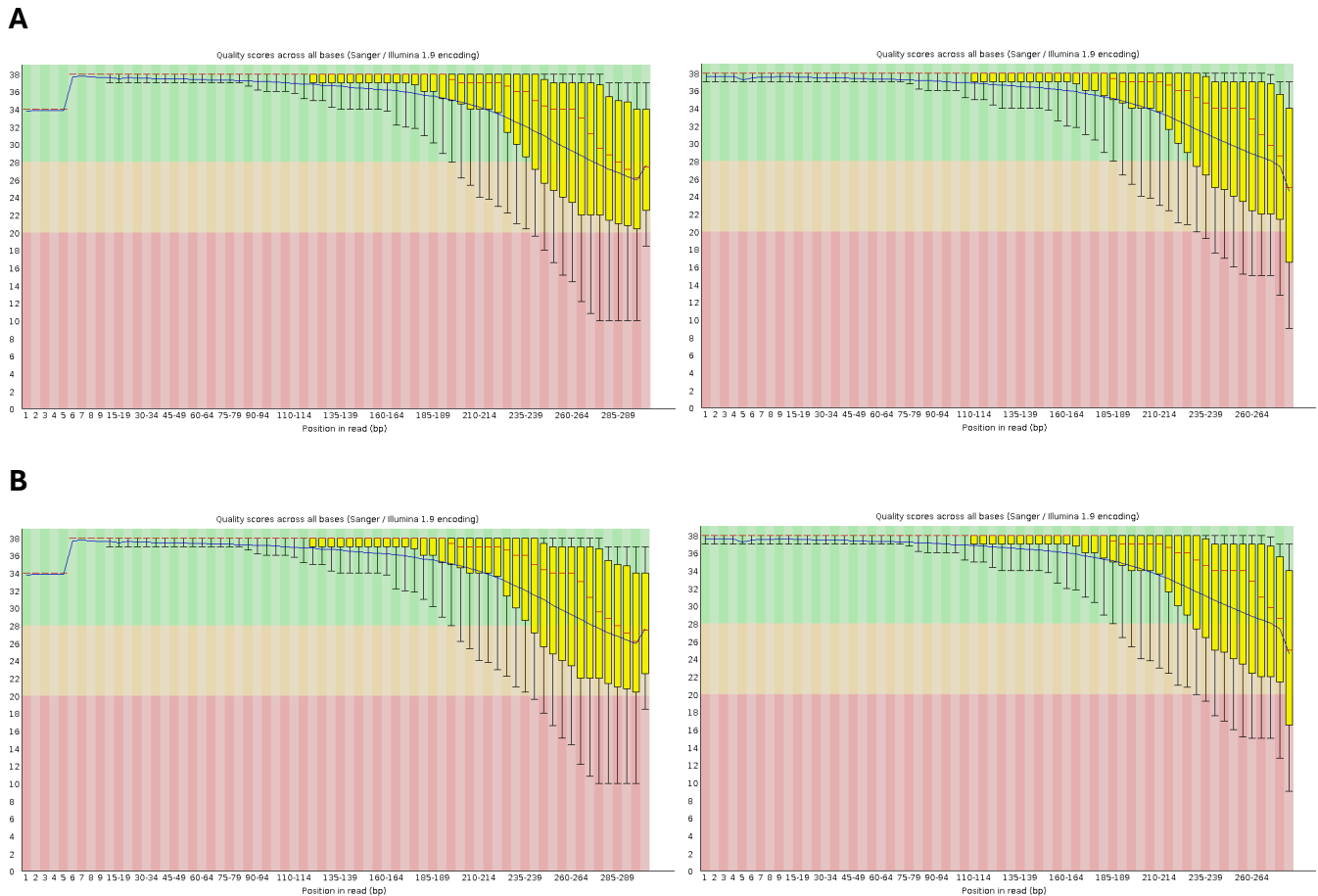


Figure 2. Post-trimming per base sequence quality. Trimmomatic (left) and fastp (right). **A)** GBG_2025_R1.fastq **B)** GBG_2025_R2.fastq

3 What is the main carbon fixation pathway used by the sequenced organism? Provide evidence from the genome in support of your answer.

To determine the main carbon fixation pathway used by the sequenced organism, structural annotation of the assembled contigs was first performed using Prokka v1.14.6 to identify potential proteins, after which functional annotation was performed using the GhostKOALA v3.1 web server to access the KEGG database and determine protein functions with associated metabolic pathways (Kanehisa *et al.*, 2016; Seemann, 2014). Using KEGG Mapper to reconstruct the pathways from the GhostKOALA results, it can be seen that 13 of the proteins identified by Prokka were mapped to KEGG Orthology (KO) identifiers associated with the Calvin cycle, while 14 were associated with various other carbon fixation pathways, indicating that the Calvin cycle is the main pathway by which carbon is fixated in this organism (Kanehisa *et al.*, 2016).

4 Does this organism encode Cytochrome C Oxidase (Cox1)? Provide the amino acid and nucleotide sequence if so.

To identify whether the sequenced organism encodes Cytochrome C Oxidase subunit 1 (Cox1), a local BLASTp search was performed with BLAST+ v2.14.1, taking the human Cox1 gene (UniProt P00395) as the query sequence and building the BLAST database from the protein FASTA file of the translated coding-sequences annotated by Prokka (Camacho *et al.*, 2009). The BLAST search confirmed that Cox1 is encoded and the Prokka ID returned was then used to retrieve the associated amino acid and nucleotide sequences (see below) from the Prokka output protein and nucleotide FASTA files.

Amino acid sequence:

MTVAISPPSKKQSDGLQPTGWLRYFSFSLDHKVIQLQYLVCGFIFYLIGGVLGGAIRIELASPIADFMGRDVYNQVLTTLHGTVMIFL
WIVPVVNGAFGNLYLIPFYVGARDMAFPRLNAVAFWMI PPAGLLLISSYFITGAAQSGWTAYPPLSITTPAAGQILWILSVLLLGGSS
IFGGINFATIILKLRQPLKLMQLPMYCWAMLGTSILVVLSTPVLAGTLILLSFDIVAHTGFFNPSLGGNAVYQHLFWFYSHPAVY
IMVLPFAFLVSEILPVHCRKPLFGYTSMVYSIMTIVILGLVVAHHMFTSGTPPMMRLFFTIATSFIAVPTGIKFFNWLATLWGGRI
SLNSAVLFSCGFIVNFVLGGITGVALAQVPFDIHVHDTYFVVAHFHYIVYGGSVFVIFASIYHWYPKFTGRMLNESLGRFHHFITFI
GFNLCFAPQHWLGLNGMPRRVAEYDPQFTVINQISSVGALLMAISTLPFLWNVVSALS GPIAGDNPWRALTPEWLTSSPPPVENWS
GKAPLVTEPYGYGVPGEELNLEKTSNSDSDLGSKSQ

Nucleotide sequence:

ATGACGGTTGCCATCTCCCTCCTAGCAAAAAGCAATCAGATGGTCTTCAACCAACTGGCTGGTTGCGTTACTTCAGCTTCAGCCTT
GACCACAAGGTCATTGGTCTGCAGTACTTGGTTTGCGGATTTCATCTTCTACCTAATCGGCGGAGTGCTTGGTGGCGCCATCCGCATC
GAACTCGCCAGCCCCATCGCCGACTTCATGGGCAGGGATGTCTACAACAGGTGCTAACGCTGCACGGCACAGTGATGATATTCCTG
TGGATCGTGCCGGTCGTGAACGGTGCTTCGGCAACTATCTCATTTCCGTTTTATGTGGGAGCCCGTGATATGGCCTTCCCTCGACTC
AATGCCGTGGCGTTCTGGATGATCCCTCCAGCTGGCCTGTTGCTGATCAGTAGCTACTTCATTACCGGTGCTGCTCAATCCGGCTGG
ACGGCCTATCCACCGCTGAGTATCACCACCCCCGCCGAGGACAGATCTTATGGATTCTCAGTGTGCTCTTGCTTGGCGGTAGCTCA
ATCTTTGGTGGCATCAACTTCATCGCCACCATCCTCAAGCTGAGGCAGCCCGGACTGAAGCTAATGCAACTACCCATGTACTGCTGG
GCGATGCTCGGCACAAGCATCCTGGTGGTGCTCTCCACACCAGTACTAGCTGGAACACTGATCCTTTTAAAGTTTCGACATCGTTGCC
CATAAGGATTCTTCAACCCAGCCTGGGCGGAAACGCAGTCGTCTACCAACATCTTTTCTGGTTTTATTTCCACCCAGCGGTATAC
ATCATGGTGTGCTGCCAGCCTTCGGATTGGTCAGCGAAATCCTTCTGTTCCTGCTGCTGCGCAAACCACTTTTTGGCTACACCTCAATGGTG
TATTCATCATGACCATCGTGATCCTTGGCCTCGTGGTTTGGGCACATCAGATGTTTCAAGTGGAACACCTCCTTGGATGCGCCTC
TTTTTACCATTGCCACATCCTTCATTGCCGTACCTACAGGAATTAAGTTCTTCAATTGGCTGGCAACACTCTGGGGAGGGCGCATC
TCTCTAAACAGTGCTGTGCTTTTCTTGTGGCTTCATTGTCAACTTTGTGCTTGGCGGCATTACTGGCGTAGCACTTGCACAAGTT
CCATTTCGACATTCACGTGCATGATACCTCTTTGTTGTTGCCCATTTTCACTACATCGTCTATGGAGGTTTCGGTGTTTGTGATCTTTG
CCTCTATTTATCACTGGTATCCAAAATTCATCTGGGCGCATGCTCAATGAATCTCTCGGGCGGTTTCATTTTTTTATCACTTTTCATCG
GCTTCAATCTTTGCTTTGCCCCACAGCATTGGCTAGGCCTCAACGGCATGCCCCGCCGTGTGGCGGAATACGACCCGCAATTCACCG
TAATCAATCAAATTAGCAGTGTTGGGGCCTTGCTAATGGCCATCAGCACGCTGCCCTTTCTCTGGAACGTCGTGCAGAGTGCATTGA
GCGGCCCCATTGCCGGAGACAACCCCTGGAGGGCGCTTACTCCTGAATGGCTCACCAGCTCCCCACCACAGTGGAACCTGGAGCG
GCAAAGCCCCCTCTGGTAACAGAGCCCTATGGCTATGGCGTACCTGGGGAGGAACCTCAACCTAGAAAAAACAGCAACAGTGATAGTG
ATCTAGGGAGCAAAAGCCAATGA

5 Retrieve homologues of Cox1 from the following representative sample of eukaryotes, Archaea, and Bacteria: *Mus musculus, *Homo sapiens**, *Drosophila melanogaster**, *Nematostella vectensis***, *Saccharomyces cerevisiae**, *Schizosaccharomyces pombe**, *Thalassiosira pseudonana****, *Arabidopsis thaliana** (eukaryotes); *Haloarcula onubensis***, *Nitrosopumilus maritimus***, *Candidatus Heimdallarchaeota archaeon LC3**** (Archaea); *Nostoc punctiforme***, *Bacillus subtilis**, *Streptomyces coelicolor**, *Pseudomonas aeruginosa**, *Rhodopseudomonas palustris***, *Bradyrhizobium japonicum*** (Bacteria).**

BLASTp was performed to identify Cox1 homologues from the above species using the UniProt webtool to search the UniProtKB database and taking the Cox1 amino acid sequence obtained from the Prokka output files as the query sequence (Bateman *et al.*, 2025). BLAST matches were selected based on lowest E-value and were first filtered to only include SwissProt sequences, as these are curated, well-annotated, and non-redundant, retrieving homologues from species marked *. Results were then filtered by unreviewed TrEMBL sequences to retrieve homologues from species marked **, as no matches were found for these in SwissProt. Of the sequences found in UniProtKB, the highest E-value was $6.3e^{-70}$, the lowest score was 582, and the lowest percentage identity was 40.2%. Homologues from the two remaining species, marked ***, could not be found in UniProtKB, so a separate search was performed with NCBI BLASTp against the NR database. Here, the highest E-value was $5e^{-113}$, the lowest query cover was 85% and the lowest percentage identity was 40.82% (Camacho *et al.*, 2009). The scoring metrics for all homologs identified were deemed acceptable and although percentage identity was low with some sequences, this is somewhat expected with distantly related species.

6 Infer a phylogenetic tree of cytochrome c oxidase. Justify the approach taken to sequence alignment and model fitting

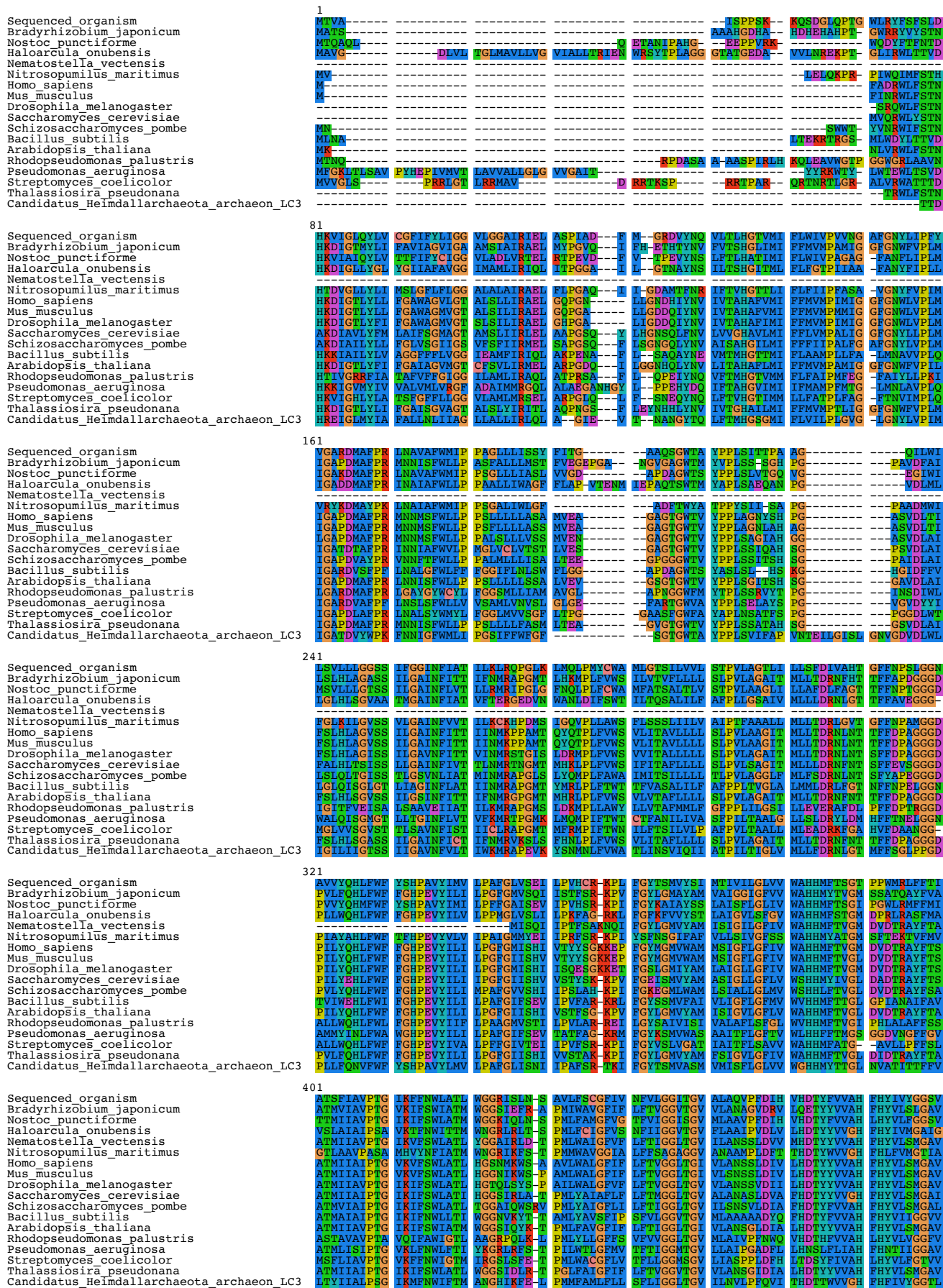
Phylogenies are constructed from an aligned set of homologous sequences so before inferring the Cox1 gene tree, a Multiple Sequence Alignment (MSA) was performed on the obtained homologous sequences using the L-INS-i algorithm from MAFFT v7.526, which is an iterative refinement method best for small-scale alignments with less than ~200 sequences. It uses a WSP score and score from local alignments during construction, allowing it to handle gaps in the alignment well (Kato and Standley, 2013). MAFFT is highly versatile, outcompeting other algorithms in most applications due to its accuracy, efficiency, and scalability (Sabonsolin and Lao, 2026). The resulting alignment was examined with Seaview v5.1 to visually check for any issues (Figure 3), of which, none were found (Gouy et al., 2021). The alignment was then used to infer a Maximum-Likelihood (ML) phylogenetic tree with IQ-TREE v3.0.1 using ModelFinder to select the best-fit model for phylogenetic analysis while support of nodes was assessed by Ultrafast Bootstrap approximation with 1000 replicates (Hoang et al., 2018; Kalyaanamoorthy et al., 2017; Wong et al., 2025). Similar to Bayesian methods, ML utilises explicit models and evaluates trees by calculating a likelihood function, making them less prone to systematic errors when compared to Parsimony methods (Kapli et al., 2020). By combining substitution models with a flexible Rate Heterogeneity Across Sites (RHAS) framework, ModelFinder has been shown to often improve fit between data, model, and tree. The optimal model can be selected for different sequence types and even to analyse distinct regions in the alignment (Kalyaanamoorthy et al., 2017). Bootstrapping is the most commonly used method for estimating confidence in a tree, with support for a tree being determined by the frequency at which the tree is inferred from pseudo-datasets (Kapli et al., 2020).

7 Plot your phylogenetic tree in a manner suitable for display in the journal Genome Biology and Evolution

To visualise the tree, the IQ-TREE output was opened in FigTree v1.1.4 and first plotted as an unrooted tree (Figure 4A), then rooted to Archaea (Figure 4B), the justification for which will be explained in question 8 (Rambaut, 2018). Taxa were coloured according to the domain of which they are a member, with green for Eukarya, blue for Bacteria, and red for Archaea. Bootstrap values were included at nodes in the rooted tree to indicate support for the tree at these points.

8 Interpret the phylogeny: Are some regions of the tree more reliable than others? Where do you think the phylogeny should be rooted, and how would different root positions alter your interpretation of the evolutionary history of the gene?

To interpret the reliability of the phylogeny, the bootstrap values were examined, the majority being found to indicate relatively high confidence with only two values below 70% (Figure 4B) (Simpson, 2010). When deciding where to root the phylogeny, it is widely agreed upon that eukaryotes evolved from an ancient Archaea that developed a number of key eukaryotic cell features, one of which was an endosymbiotic relationship with a Bacteria (Kay et al., 2026). Therefore, it follows that the Cox1 gene in eukaryotes must either be derived from Archaea or Bacteria, so one of these should be used as the outgroup. With Bacteria as the outgroup, the implication is that the gene evolved from the archaeal version which is interesting given the potential origin of eukaryotes within ancient Asgardarchaeota, of which *Candidatus H. archaeon LC3* is a member (Zhang et al., 2025). However, aerobic respiration in eukaryotes occurs within the mitochondria, meaning that it is more likely that the gene evolved from that early endosymbiotic Bacteria, making it more logical to place Archaea as the outgroup (Kay et al., 2026).





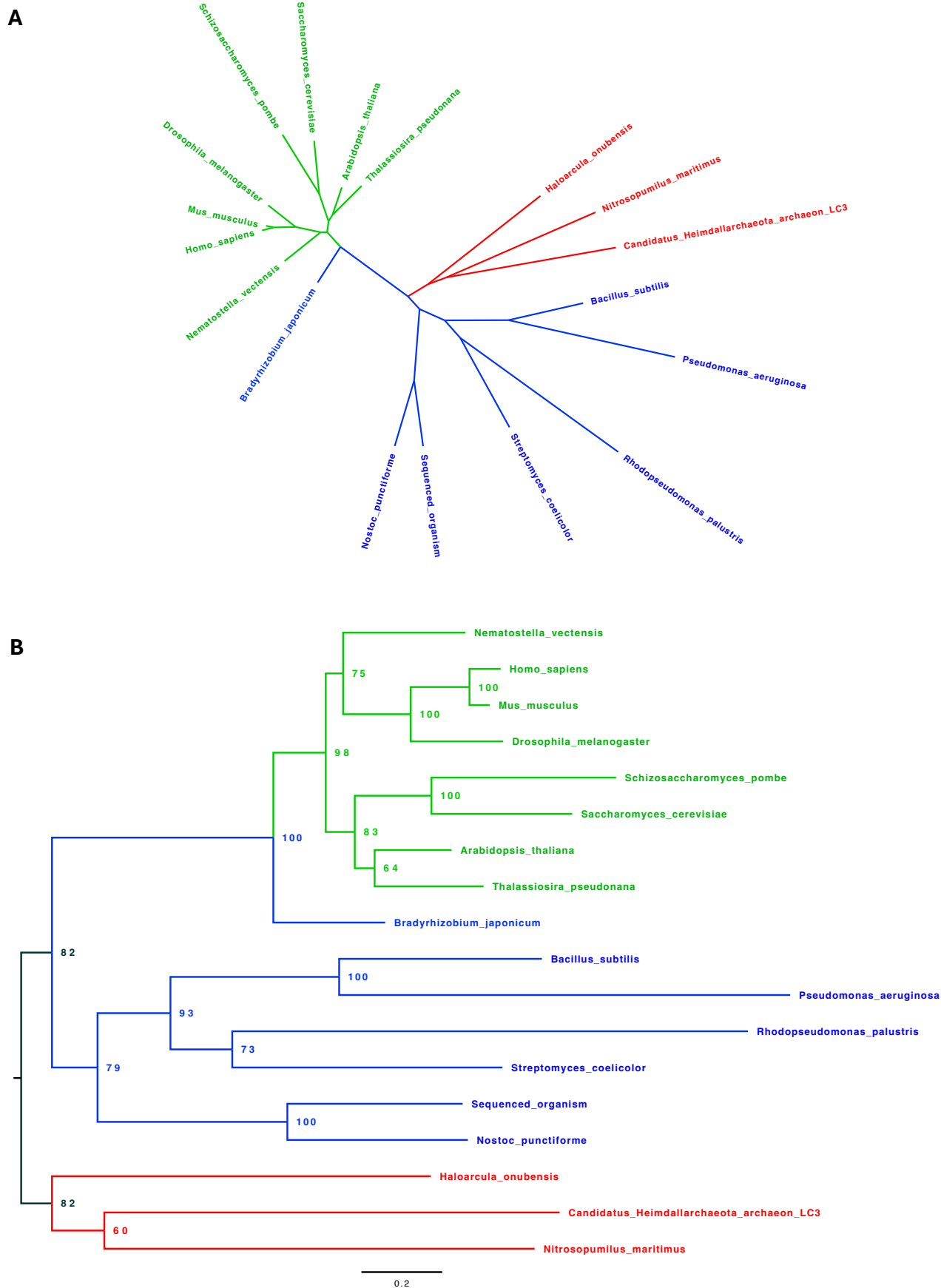


Figure 4. Cox1 phylogenetic trees. Eukaryotes (green), bacteria (blue), and archaea (red). **A)** Unrooted **B)** Rooted.

9 Do the eukaryotic sequences branch where you expect in the tree? Why or why not, and how do you interpret their inferred phylogenetic position?

The eukaryotic sequences mostly branch as expected with one notable exception being the branching of the sequences belonging to *A. thaliana* and *T. pseudonana* which should form their own clade earliest in the tree but are instead found within the same clade as those of *S. cerevisiae* and *S. pombe*, which should branch off next. However, beyond this, the topology is broadly consistent with expectations: The *N. vectensis* sequence diverges next, followed by that of *D. melanogaster*, leaving those of *H. sapiens* and *M. musculus* to form their own clade. The eukaryotes as a whole diverge from Bacteria, with *B. japonicum* being their closest bacterial relative. This makes sense given that *B. japonicum* is an Alphaproteobacteria and that the endosymbiont from which the Cox1 gene in eukaryotes likely evolved is believed to have been Alphaproteobacteria-like and an ancestor of extant lineages (Fan et al., 2020; Ong and O'Brian, 2024). The expected positions outlined were determined using the OneZoom Tree of Life (OneZoom Core Team, 2026; Wong and Rosindell, 2022). It should be noted that a limitation of this approach is that this webtool is essentially a species tree and will not necessarily reflect the evolution of specific genes.

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